Homework 2

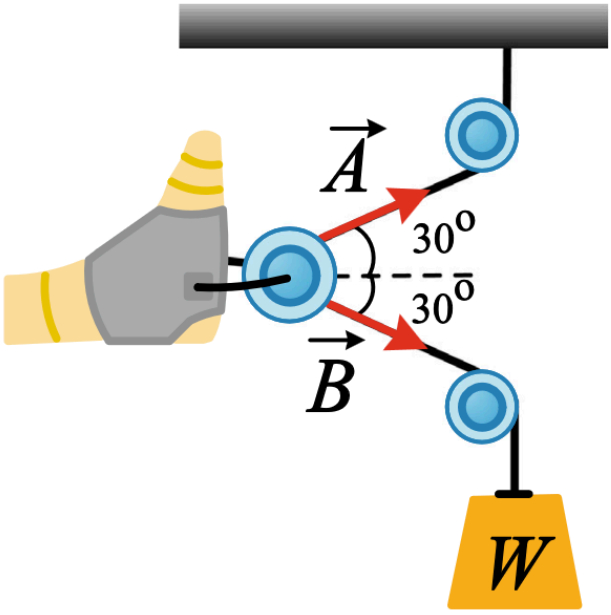
Due Tuesday, September 8, 2026 by 11:59 PM on Gradescope

Instructions

- Include your full name and Coyote ID on the top right of the first page
- Clearly circle or box your answer for any question involving calculations. If the problem is multipart, circle or box answers for each part.
- Work should be neatly written and easy to follow.
- You must tag each question on Gradescope. If you do not tag your work for a given problem, you will not receive credit for it.

Problems

This homework contains some vector decomposition problems (material from week 1), motion graphs, as well as average velocity and and average acceleration in one-dimension.



1. The figure above show s a traction apparatus which is used to prevent muscle contraction on a patient's leg. The pull of the two cords, which are presented by the vectors $\vec{A}$ and $\vec{B}$, have equal magnitude and combine to produce an outward horizontal traction force of $10\,\mathrm{N}$ on the patient's leg. Find the values of each pull (i.e., $\left|\vec{A}\right|$ and $\left|\vec{B}\right|$).

"$\vec{A}$ and $\vec{B}$, have equal magnitude and combine to produce an outward horizontal traction force of $10\,\mathrm{N}$" means the horizontal direction of one of the triangles is $5\,\mathrm{N}$. Using $\cos\theta = \frac{\text{adj}}{\text{hyp}}$ $A = B = \frac{5\,\mathrm{N}}{\cos 30^\circ} = 5.77\,\mathrm{N}$

2. Convert the following to basic SI units or a combination of basic SI units (i.e., kilograms, meters, and seconds – kg, m, and s):

   (a) $8.0\,\text{in}$ $0.2032\,\text{m}$

   (b) $66\,\text{ft/s}$ $20.12\,\text{m/s}$

   (c) $60\,\text{mph}$ $26.82\,\text{m/s}$

   (d) $14\,\text{in}^2$

$$14\,\text{in}^2 \cdot \left(\frac{1\,\text{m}}{39.37\,\text{in}}\right)^2 = 14 \cdot \left(\frac{1\,\text{m}^2}{1550}\right) = 0.009032\,\text{m}^2$$

3. A runner on a straight track covers $1\,\text{km}$ in $4\,\text{min}$. What is their average velocity in

   (a) km/min $0.25\,\text{km/min}$

   (b) km/s $0.004167\,\text{km/s}$

   (c) km/h $15\,\text{km/h}$

4. Perform the following calculations with the correct number of significant figures:

   (a) 159.31×204.6 32.60×10^3

   (b) $5.1125 + 0.67 + 3.2$ 9.0

   (c) $7.662 - 7.425$ 0.237

   (d) $16.5/3.45$ 4.78

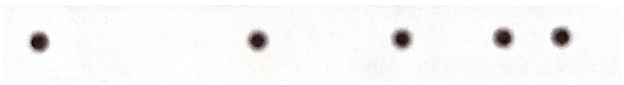
5. The quantity called *mass density* is the mass per unit volume of a substance. What are the mass densities in basic SI units (i.e., kilograms, meters, and seconds – kg, m, and s) of the following objects:

   (a) $215\,\mathrm{cm}^3$ solid with a mass of $0.0179\,\mathrm{kg}$.

$83.26 \times 10^{-6}\mathrm{kg/cm}^3 = 83.26\mathrm{kg/cm}^3$

   (b) $95\,\mathrm{cm}^3$ liquid with a mass of $77\,\mathrm{g}$.

$0.81\,\mathrm{g/cm}^3 = 810\,\mathrm{kg/m}^3$



Is the particle in the figure above speeding up? Slowing down? Or can you tell? Explain.

Without numbers on the dots we cannot tell if the particle in the figure is moving left or right, so we can't tell if it is speeding up or slowing down. If the particle is moving to the right, it is speeding up. If it is moving to the left, it is slowing down.



6. Does the object represented in the figure above have a positive or negative value of a_x? Explain.

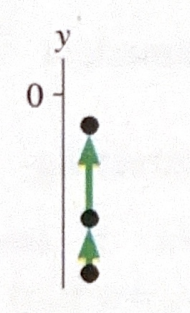
Because the velocity vectors get shorter for each time step, the object must be slowing down as it travels to the left. The acceleration vector must therefore point in the opposite direction as the velocity, so the acceleration vector points to the right. Thus, a_x is positive as per our convention (see Tactics Box 1.3).



7. Does the object represented in the figure above have positive or negative value of a_y? Explain.

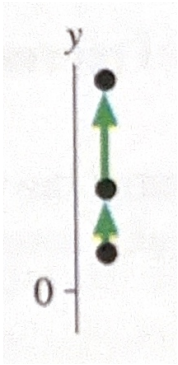
Because the velocity vectors get shorter for each time step, the object must be slowing down as it travels in the $-y$ direction (down). The acceleration vector must therefore point in the direction opposite to the velocity; namely, in the $+y$ direction (up). Thus, a_y is positive as per our convention (see Tactics Box 1.3).

8. Determine the signs (positive, negative, or zero) of the position,



velocity, and acceleration for the particle in the figure above.

The particle position is below zero on the y-axis, so its position is negative. The particle is moving up, so its velocity is positive. The particle's speed is increasing as it moves in the positive direction, so its acceleration vector points in the same direction as its velocity vector (i.e., up). Thus, the acceleration is also positive.

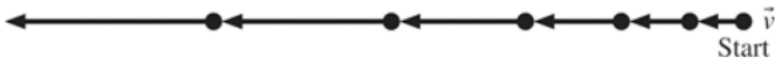


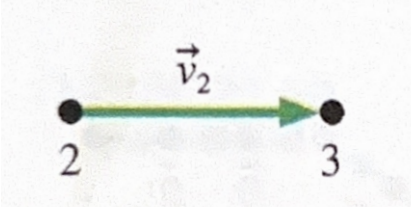
9. Determine the signs (positive, negative, or zero) of the position, velocity, and acceleration for the particle in the figure above.

The particle position is above zero on the y-axis, so its position is positive. The particle is moving up, so its velocity is positive. The particle's speed is increasing as it moves in the positive direction, so its acceleration vector points in the same direction as its velocity vector (i.e., up). Thus, the acceleration is also positive.

10. A baseball player stats running to the left to catch the ball as soon as the hit is made. Use the particle model to draw a motion diagram showing the position and average velocity vectors of the player during the first few seconds of the run.

SOLVE: The player starts from rest and moves faster and faster to the left.





11. The figure above shows two dots of a motion diagram and vector $\vec{v}_2$.

Copy this figure, then add dot 4 and the next velocity vector $\vec{v}_3$:

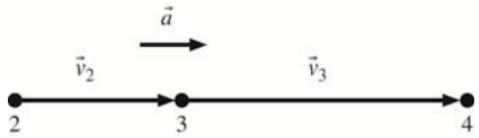
(a) if the acceleration vector $\vec{a}$ at dot 3 points **right**

(b) if the acceleration vector $\vec{a}$ at dot 3 points **left**

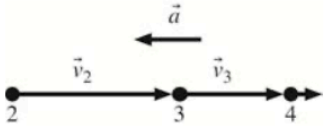
VISUALIZE: If the acceleration points to the right, the rightward velocity vector should become longer. If the acceleration points to the left, the rightward velocity vector should become shorter.

SOLVE:

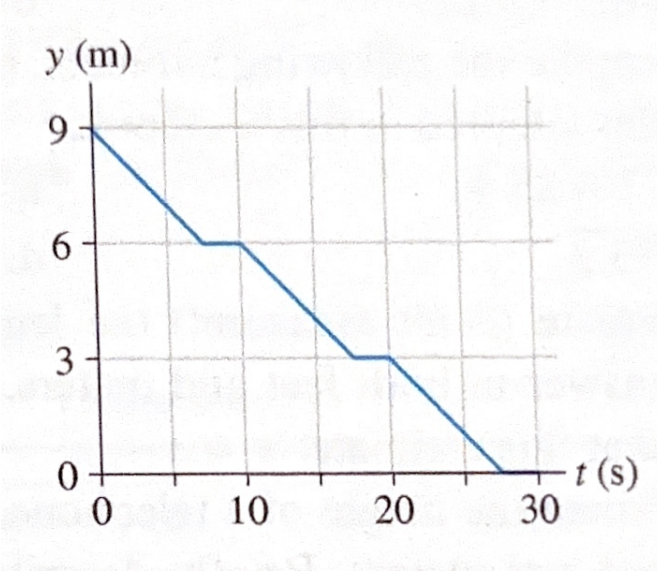
(a)



(b)



REVIEW: Any time the acceleration points in the same direction as the velocity, the size of the velocity vector will increase, as in part **(a)**.



12. Write a short description of the motion of a real object for which the figure above would be a realistic position versus time graph.

MODEL: We will treat the object in our description as though it has a single point-like position (the particle model).

VISUALIZE: We must use an example that is moving in the vertical direction, downward, with pauses of 2-3 seconds in between periods of motion. The speed is approximately 0.5 m/s.

SOLVE: Many answers may be possible. A firefighter at the end of a motorized latter is being moved from the top of a three-story building down to the ground. She starts at a height of 9 m, and is lowered 3 m to the next floor in 6 s. At that floor, she pauses for 4 s, peering into a window to make sure all fire has been extinguished. She is then lowered another 3 m in 6 s. Again, she pauses for 4 s to peer inside the building. Finally, she is lowered the final 3 m to the ground, in 6 s.

REVIEW: Our example addresses the starting and ending positions, the times involved, and the rate of change of the position at various times.

13. A car travels in the positive x-direction for $20\,\text{km}$ at $40\,\text{km/h}$. It then continues in the same direction for another $20\,\text{km}$ at $80\,\text{km/h}$.

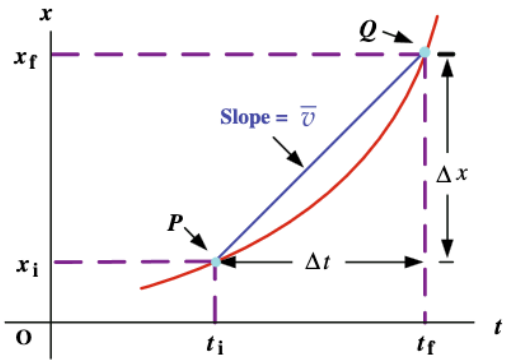
(a) What is the average velocity of the car during this $40\,\text{km}$ trip?

$$t_1 = \frac{20\,\text{km}}{40\frac{\text{km}}{\text{h}}} = 0.5\,\text{h},\ t_1 = \frac{20\,\text{km}}{80\frac{\text{km}}{\text{h}}} = 0.15\,\text{h},$$

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{20\,\text{km} + 20\,\text{km}}{0.5\,\text{h} + 0.25\,\text{h}} = 53.33\,\text{km/h}$$

(b) What is the average speed?

Because both velocities are in the same direction and the direction is positive the average speed is equal to the average velocity.



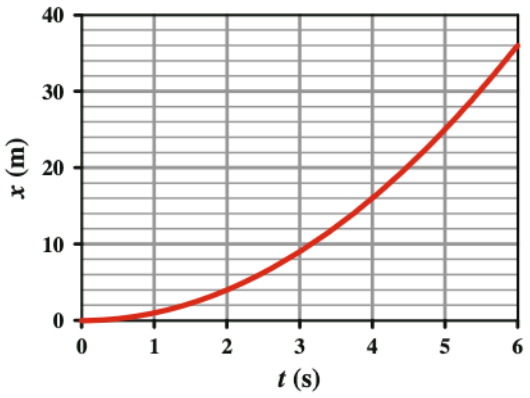
14. Suppose the motion of the particle in the above figure is described by the equation $x = a + bt^2$, where $a = 10\,\text{m}$ and $b = 2\,\text{m/s}^2$.

(a) Find the displacement of the particle in the time interval between $t_i = 2\,\text{s}$ and $t_f = 4\,\text{s}$.

$x_i = 18\,\text{m}, x_f = 42\,\text{m}, \Delta x = x_f - x_i = 24\,\text{m}$

(b) Find the average velocity and the average speed during this interval of time.

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{24\,\text{m}}{2\,\text{s}} = 12\,\text{m/s}, \text{ speed} = |v_{\text{avg}}| = |12\,\text{m/s}| = 12\,\text{m/s}$$



15. A graph of position (in meters) versus time (in seconds) for a person traveling in the positive x-direction is displayed in the figure above. Find the average velocity for the following cases:

(a) $t_i = 2\,\text{s}$ and $t_f = 4\,\text{s}$

$\Delta t = t_f - t_i = 2\,\text{s}$, from the plot $x_i = 4$ and $x_f = 16$,

$$\Delta x = x_f - x_i = 12\,\text{m},\ v_{\text{avg}} = \frac{\Delta x}{\Delta t} = 6\,\text{m/s}$$

(b) $t_i = 2\,\text{s}$ and $t_f = 6\,\text{s}$

$$\Delta t = 4\,\text{s},\ \Delta x = 32\,\text{m},\ v_{\text{avg}} = \frac{\Delta x}{\Delta t} = 8\,\text{m/s}$$

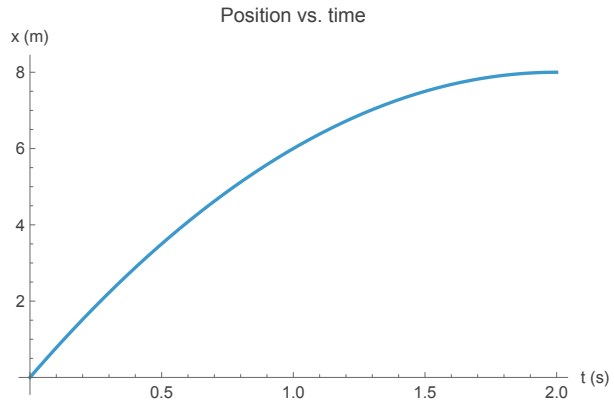
(c) $x_i = 12\,\text{m}$ and $x_f = 30\,\text{m}$

From the plot $t_i \approx 3.5\,\text{s}$ and $t_f \approx 5.5\,\text{s}$, $\Delta t = 2\,\text{s}$, $\Delta x = 18\,\text{m}$,

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = 9\,\text{m/s}$$

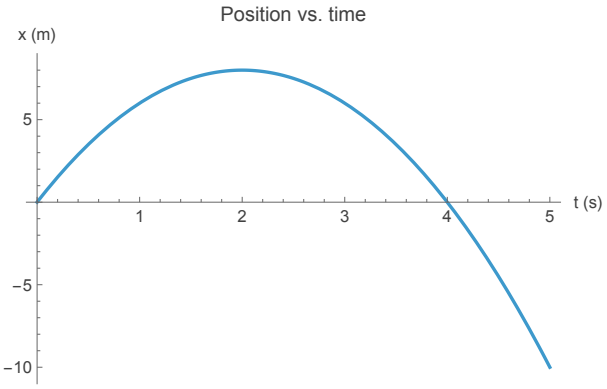
16. A body moves along a straight line with position even by $x = 8t - 2t^2$, where x is in meters and t is in seconds. Fine the average velocity and average speed of the body in the intervals:

(a) From $t_i = 0$ to $t_f = 2\,\text{s}$



$x_i = 0\,\text{m}, x_f = 8\,\text{m}, \Delta x = 8\,\text{m}, \Delta t = 2\,\text{s}, v_{\text{avg}} = \dfrac{\Delta x}{\Delta t} = 4\,\text{m/s}$, because the velocity is always positive the average speed is equal to the average velocity.

(b) From $t_i = 0$ to $t_f = 5\,\text{s}$



$x_i = 0\,\text{m}, x_f = -10\,\text{m}, \Delta x = -10\,\text{m}, \Delta t = 5\,\text{s}, v_{\text{avg}} = \dfrac{\Delta x}{\Delta t} = -2\,\text{m/s}$

Because velocity changes from positive to negative, we have to find the turning point of the position graph (the point where the tangent of the position graph changes direction) you could use calculus to do this, a plot, or a table of values, anyway you do it you should get the turning point to be $t_{\text{turn}} = 2\,\text{s}$. Recall that average speed is always positive so $x_{\text{turn}} = 8\,\text{m}$, $\Delta x_{0\to\text{turn}} = 8\,\text{m}$, $\Delta t_{0\to\text{turn}} = 2\,\text{s}$, $|v_{\text{avg}}| = 4\,\text{m/s}$, $\Delta x_{\text{turn}\to 5} = -18\,\text{m}$, $\Delta t_{\text{turn}\to 5} = 3\,\text{s}$, $|v_{\text{avg}}| = 6\,\text{m/s}$, the average speed is the weighted average of both speeds

$$\text{speed}_{\text{avg}} = \frac{2}{5}\cdot 4\frac{\text{m}}{\text{s}} + \frac{3}{5}\cdot 6\frac{\text{m}}{\text{s}} = 5.2\,\text{m/s}$$